Ex. No. 2	**Adaptive and Responsive UI Design**
Date of Exercise	
Github Code Link (Keep as public)	
Video Link (Google Drive Link)	

Aim

Design and implement a responsive Flutter application for an E-Commerce Product Catalog by creating reusable custom product widgets. Ensure the application adapts to different screen sizes using responsive layout techniques.

Description

Responsive design in Flutter means the same widget tree rearranges itself to make good use of whatever space is available, instead of assuming a fixed phone-sized screen. LayoutBuilder gives a widget access to the BoxConstraints handed down by its parent, so the number of columns in a grid, the padding, or even which widgets are shown can be computed from the actual available width at build time.

This experiment implements an E-Commerce Product Catalog. A Product model stores the name, category, price, star rating, an emoji icon and an accent colour for each item. A reusable ProductCard widget composes the thumbnail, name, category, rating row and a price/add-to-cart row inside a Card. The CatalogScreen wraps a GridView.builder in a LayoutBuilder and picks the crossAxisCount from the current width: 2 columns under 600px, 3 under 900px, 4 under 1200px, and 5 beyond that. A banner at the top reports the live column count and width so the responsive behaviour is visible while resizing. Running the same build at a narrow and a wide viewport proves the layout adapts without any platform-specific code.

Program

lib/models/product.dart

```
import 'package:flutter/material.dart';

class Product {
  final String name;
  final String category;
  final String emoji;
  final double price;
  final double rating;
  final Color color;
```

```
  const Product({
    required this.name,
    required this.category,
    required this.emoji,
    required this.price,
    required this.rating,
    required this.color,
  });
}

const List<Product> products = [
  Product(name: 'Wireless Headphones', category: 'Audio', emoji: '🎧', price:
2499, rating: 4.5, color: Color(0xFF42A5F5)),
  Product(name: 'Smart Watch', category: 'Wearables', emoji: '⌚', price: 3999,
rating: 4.2, color: Color(0xFF66BB6A)),
  Product(name: 'Running Shoes', category: 'Footwear', emoji: '👟', price:
1899, rating: 4.6, color: Color(0xFFFFA726)),
  Product(name: 'Coffee Maker', category: 'Home', emoji: '☕', price: 3299,
rating: 4.1, color: Color(0xFF8D6E63)),
  Product(name: 'Backpack', category: 'Accessories', emoji: '🎒', price: 1499,
rating: 4.4, color: Color(0xFFAB47BC)),
  Product(name: 'Bluetooth Speaker', category: 'Audio', emoji: '🔊', price:
1799, rating: 4.3, color: Color(0xFF26C6DA)),
  Product(name: 'Desk Lamp', category: 'Home', emoji: '💡', price: 899, rating:
4.0, color: Color(0xFFFFEE58)),
  Product(name: 'Sunglasses', category: 'Accessories', emoji: '🕶️', price:
1299, rating: 4.3, color: Color(0xFF5C6BC0)),
  Product(name: 'Yoga Mat', category: 'Fitness', emoji: '🧘', price: 999,
rating: 4.5, color: Color(0xFFEF5350)),
  Product(name: 'Water Bottle', category: 'Fitness', emoji: '🚰', price: 499,
rating: 4.7, color: Color(0xFF29B6F6)),
  Product(name: 'Keyboard', category: 'Electronics', emoji: '⌨️', price: 2199,
rating: 4.4, color: Color(0xFF78909C)),
  Product(name: 'Desk Chair', category: 'Home', emoji: '🪑', price: 6499,
rating: 4.2, color: Color(0xFF8D6E63)),
];
```

lib/widgets/product_card.dart

```
import 'package:flutter/material.dart';

import '../models/product.dart';

/// Reusable product card used by the responsive catalog grid.
class ProductCard extends StatelessWidget {
  final Product product;

  const ProductCard({super.key, required this.product});

  @override
  Widget build(BuildContext context) {
    return Card(
      elevation: 2,
      shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(16)),
      clipBehavior: Clip.antiAlias,
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.stretch,
        children: [
          Container(
            height: 90,
            color: product.color.withValues(alpha: 0.2),
            alignment: Alignment.center,
            child: Text(product.emoji, style: const TextStyle(fontSize: 42)),
          ),
          Padding(
            padding: const EdgeInsets.fromLTRB(10, 8, 10, 10),
            child: Column(
              crossAxisAlignment: CrossAxisAlignment.start,
              children: [
                Text(
                  product.name,
```

```
                  maxLines: 1,
                  overflow: TextOverflow.ellipsis,
                  style: const TextStyle(fontWeight: FontWeight.bold, fontSize:
14),
                ),
                const SizedBox(height: 2),
                Text(
                  product.category,
                  style: TextStyle(fontSize: 11, color: Colors.grey.shade600),
                ),
                const SizedBox(height: 4),
                Row(
                  children: [
                    const Icon(Icons.star, size: 14, color: Colors.amber),
                    const SizedBox(width: 2),
                    Text(product.rating.toStringAsFixed(1), style: const
TextStyle(fontSize: 12)),
                  ],
                ),
                const SizedBox(height: 6),
                Row(
                  mainAxisAlignment: MainAxisAlignment.spaceBetween,
                  children: [
                    Text(
                      '₹${product.price.toStringAsFixed(0)}',
                      style: TextStyle(fontWeight: FontWeight.bold, fontSize:
14, color: product.color),
                    ),
                    InkWell(
                      borderRadius: BorderRadius.circular(20),
                      onTap: () {},
                      child: Container(
                        padding: const EdgeInsets.all(6),
                        decoration: BoxDecoration(color: product.color, shape:
BoxShape.circle),
                        child: const Icon(Icons.add_shopping_cart, size: 14,
color: Colors.white),
                      ),
                    ),
                  ],
                ),
              ],
            ),
          ),
        ],
      ),
    );
  }
}
```

lib/screens/catalog_screen.dart

```
import 'package:flutter/material.dart';

import '../models/product.dart';
import '../widgets/product_card.dart';

class CatalogScreen extends StatelessWidget {
  const CatalogScreen({super.key});

  int _columnsForWidth(double width) {
    if (width < 600) return 2;
    if (width < 900) return 3;
    if (width < 1200) return 4;
    return 5;
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Product Catalog'), centerTitle: true),
```

```
      body: LayoutBuilder(
        builder: (context, constraints) {
          final columns = _columnsForWidth(constraints.maxWidth);
          return Column(
            children: [
              Container(
                width: double.infinity,
                padding: const EdgeInsets.all(12),
                color: Theme.of(context).colorScheme.secondaryContainer,
                child: Text(
                  'Showing $columns column${columns == 1 ? '' : 's'} at
${constraints.maxWidth.toStringAsFixed(0)}px width',
                  style: const TextStyle(fontSize: 13, fontWeight:
FontWeight.w600),
                ),
              ),
              Expanded(
                child: GridView.builder(
                  padding: const EdgeInsets.all(12),
                  gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(
                    crossAxisCount: columns,
                    mainAxisSpacing: 12,
                    crossAxisSpacing: 12,
                    childAspectRatio: 0.72,
                  ),
                  itemCount: products.length,
                  itemBuilder: (context, index) => ProductCard(product:
products[index]),
                ),
              ),
            ],
          );
        },
      ),
    );
  }
}
```

lib/main.dart

```
import 'package:flutter/material.dart';

import 'screens/catalog_screen.dart';

void main() {
  runApp(const CatalogApp());
}

class CatalogApp extends StatelessWidget {
  const CatalogApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Product Catalog',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.indigo),
        useMaterial3: true,
      ),
      home: const CatalogScreen(),
    );
  }
}
```

Output Screenshots

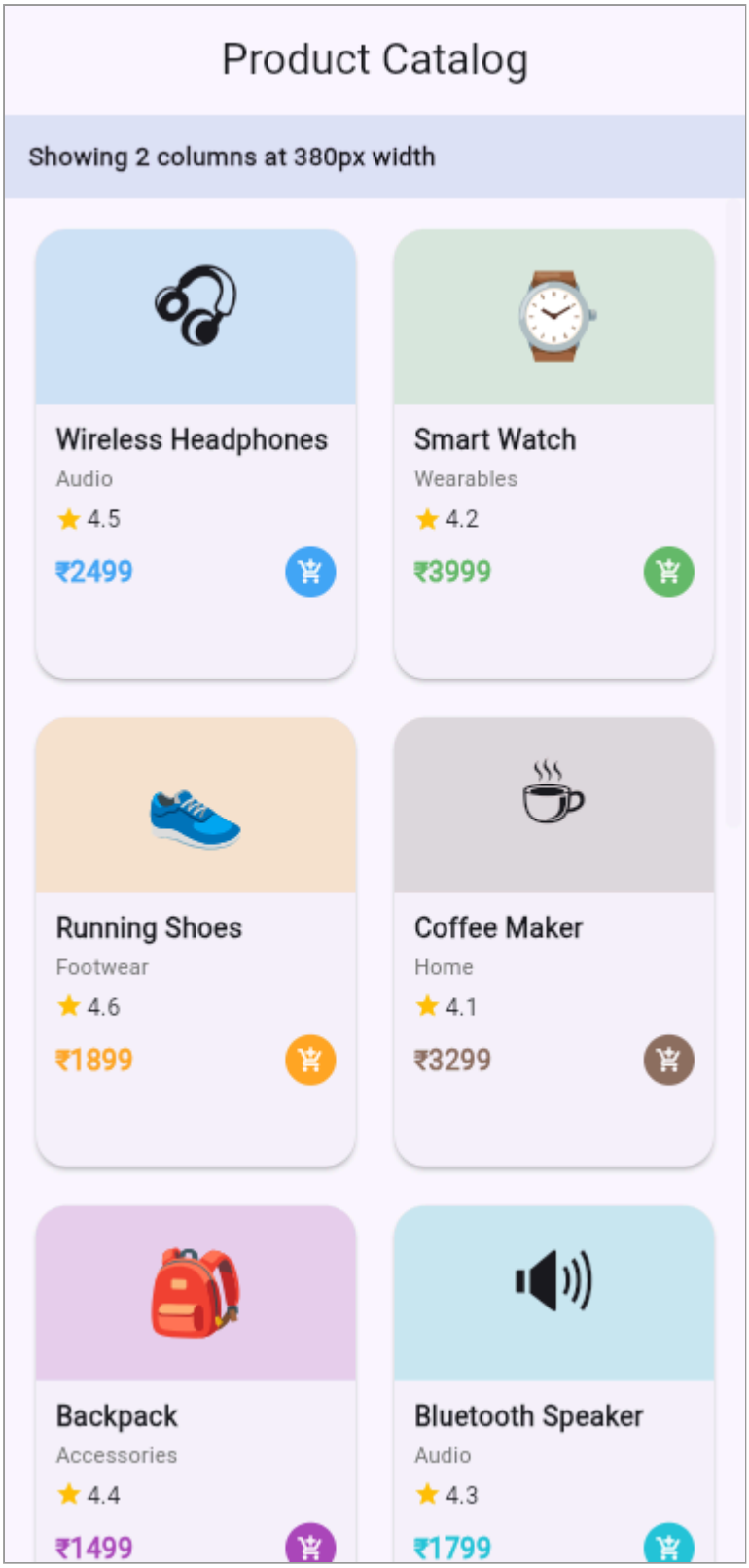


Fig 1. At 380px width the responsive grid renders 2 columns.

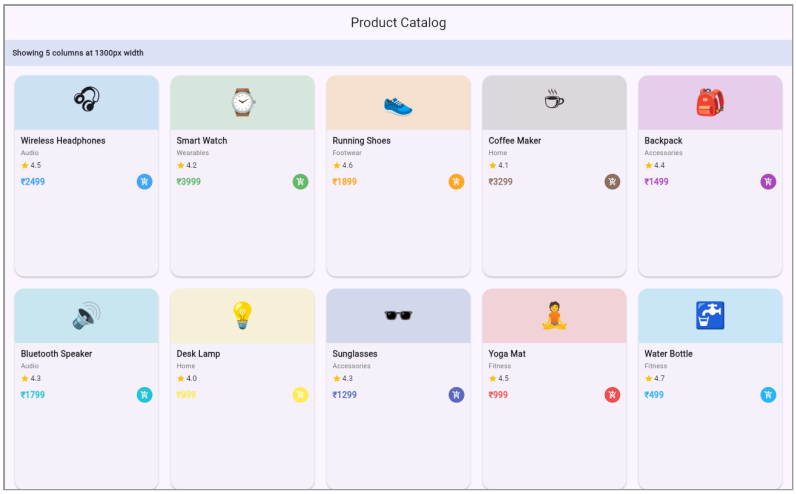


Fig 2. The same CatalogScreen at 1300px width automatically renders 5 columns, proving the LayoutBuilder-driven responsive behaviour.

Result

The responsive E-Commerce Product Catalog was successfully designed and implemented using a reusable ProductCard widget and a LayoutBuilder-driven GridView. The application was verified to adapt its column count correctly across narrow and wide screen widths, as shown in the screenshots above.